Can a VLM Grade the Robot? Vision-Language Models as Reward Models for Self-Improving Robot Policies

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1 Motivation / Problem Statement

Can robots learn only from their experience?

A lot of the focus in machine learning rests on supervised and self-supervised learning over enormous human-generated datasets. This recipe works amazingly wherever such data is abundant, but axiomatically stalls wherever gold labels do not exist or are prohibitively expensive. Robotics is a clear example of this: every demonstration must be physically produced, usually by a human teleoperator, and no dataset can anticipate every home, factory, or object a robot will encounter [1, 2]. The Big World Hypothesis [3] makes this point in a more general form: the world is vastly larger than any agent, so no amount of prior training can fully prepare them. A robot working in a new domain will have to keep learning from its own experience, much as living intelligence does.

In this project, we aim to test how this bottleneck can be bypassed. To that respect, granting to the models the capability to judge their own actions, and subsequently learn from them, is a crucial step. This is precisely what we want to examine in this project.

- **Implementation:** How do existing solutions utilize VLMs as reward models [4–6], and what are their specific technical approaches?
- **Performance:** What results do they achieve, and how they compare to other learning methods?
- **Bottlenecks:** What is their current state, and what are their primary limitations?

What validates our approach, and the core idea behind it, is that one may be able to understand when a task is completed, and how well it was performed, but not know how to achieve it. Reinforcement learning [7] turns this asymmetry into a concrete, principled learning approach, but it shifts the burden onto the reward function, which in robotics has traditionally been hand-engineered per task with custom sensors or instrumented environments. Vision-language models offer a method that circumvent this as a manifestation of the Bitter Lesson [8]: given only a language description of the task and the robot's camera observations, a VLM should be able to judge success and progress, acting as a general-purpose reward model [4–6].

We see this as a natural direction that the field will eventually move to, and recent work already improves vision-language-action policies from their own experience [9, 10]. As existing sources of human-generated data become exhausted, experience becomes the largest untapped source left, and figuring out how to let the model judge its actions is a crucial step to that, with VLMs being a valuable generalist foundation for that potentially. We think that this project is a great opportunity to get involved with this direction and dive into the respective literature.

2 Proposed Approach

We use rejection sampling fine-tuning [11] on $\pi_{0.5}$ [1] in LIBERO [12]: roll out the policy, keep the episodes judged successful, LoRA [13] fine-tune on them, repeat. The one place error can enter is the pass/fail label on each episode, which is the variable we study.

Stage	Description
Pipeline	Get the $\pi_{0.5}$-base/$\pi_{0.5}$-LIBERO checkpoints [14] running in the LIBERO simulator [12] on the H100. Build rollout and logging infrastructure that stores frames, actions, and the success flag per episode. Figure out compute, set up the environments and the algorithms. Potentially finetune it on our dataset.
Sanity gate & grader audit	Run the oracle-labeled loop for a few rounds (2–3 rounds). Score every rollout and build a confusion matrix against the oracle.
VLM-labeled loop	Swap the simulator's success flag for a VLM grader (RoboReward [6], VLM-RM [4], GVL [5], etc.) and see how well it matches the ground-truth training run. Compare different approaches of how to deploy such training schemes.

Table 1: Project stages.

3 Hypotheses & Initial Experiments

We are trying to figure out what the bottleneck is in using VLMs as reward models. In that, we want to see how performance of different such approaches compares to baselines, what weaknesses/strenghts are highlighted during training and testing, and what questions remain open.

Expectations

- **Limitations vs Potential:** We expect VLMs to introduce significant noise and false positives that could hinder learning. However, freeing the loop from gold-label requirements allows us to utilize vastly more data, which may circumvent this noise or grant unforeseen advantages.
- **Credit Assignment & Reward Hacking:** We want to observe first-hand how the credit assignment problem [7] plays out and whether reward hacking (the RL algorithm finding and exploiting gaps in the VLM's value function) [15] manifests itself during our experiments.

Evaluation

- **Method Comparison:** By comparing different VLM reward models [4–6], we hope to identify their main advantages, disadvantages, and structural bottlenecks. Deep understanding of the literature is an important step we hope to achieve here.
- **Evaluation Axes:** After training using selected VLMs on unified benchmarks, we will compare them to supervised fine-tuned models (like openpi's [14] or our own version) across several axes: success rate, recall, training time, and qualitative differences in the policies.
- **Label-Free Domains:** An interesting direction would be to test the VLM reward scheme in a domain completely lacking gold labels. Achieving an acceptable success rate in such scenarios would strongly validate this approach as a principled solution, given that there are limited solutions for what we can do in such scenarios.
- **Custom Variations (Perhaps):** If project limits allow and we come up with a good idea on top of the existing literature, we may develop our own variation of a VLM reward model.

4 Resources

Compute.

- PACE (H100).
- RTX 5090, 128 GB RAM workstation (Stratos): development, ablation.

Project GitHub. `https://github.com/EthanMBoos/self-improving-policies`

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